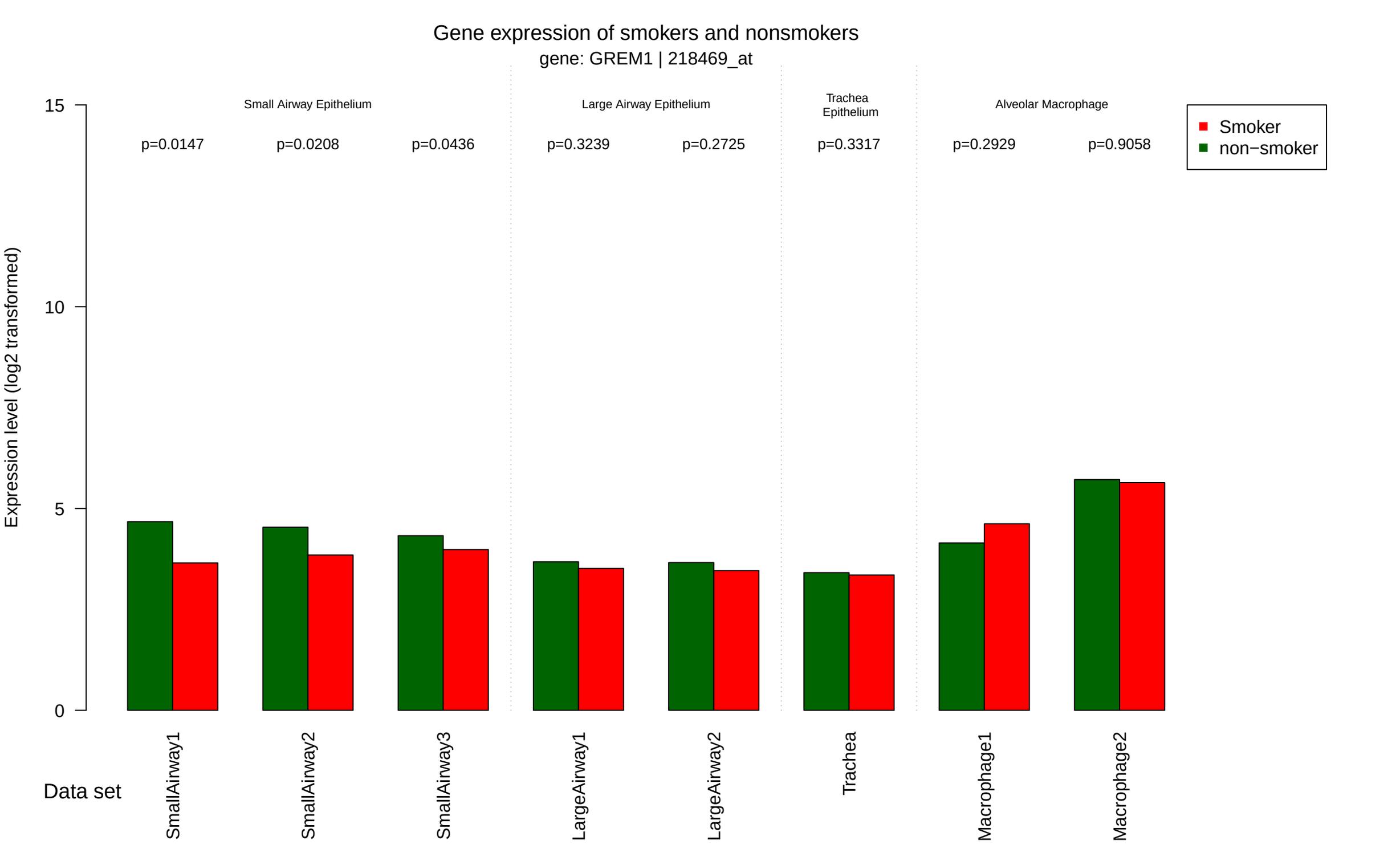
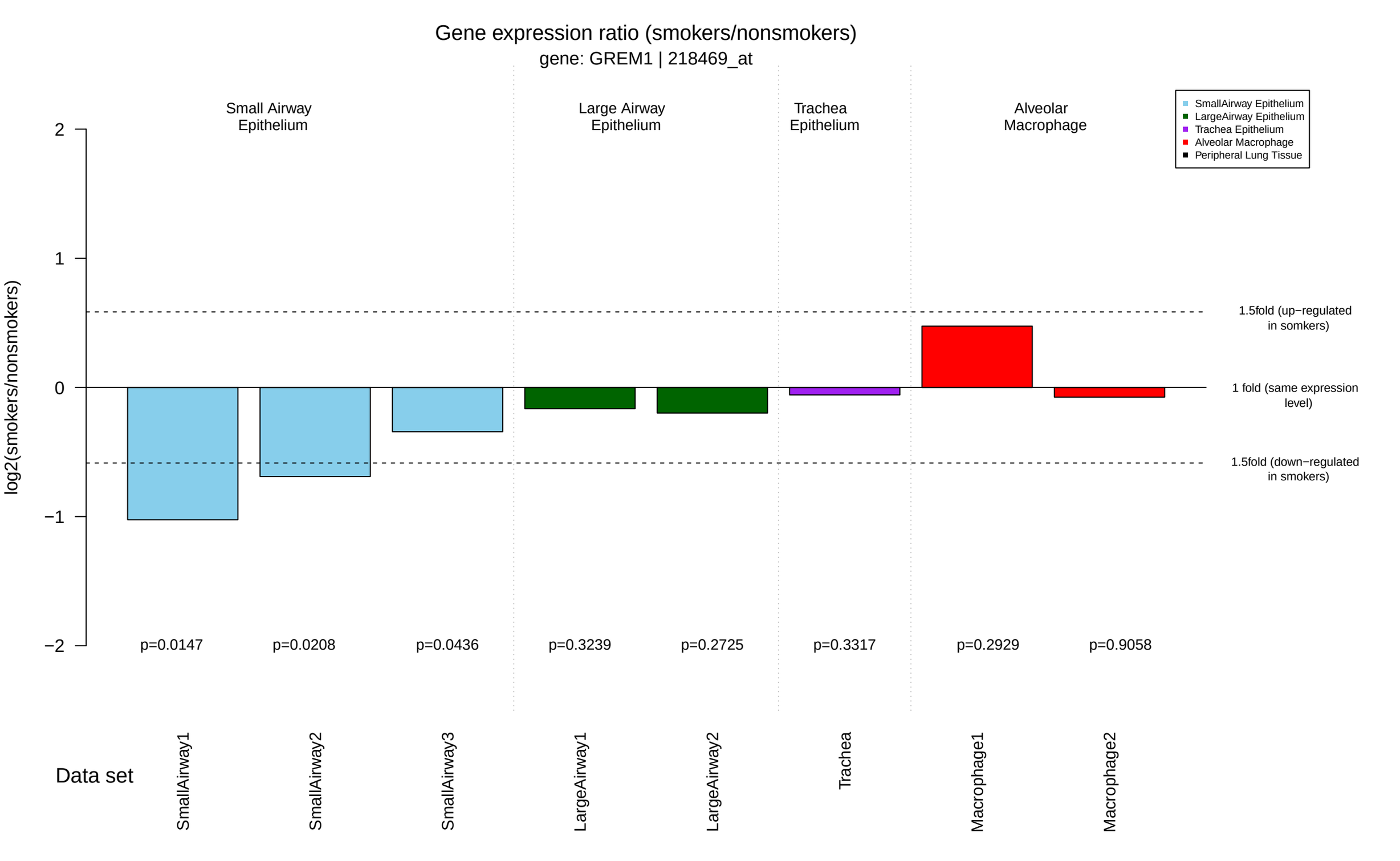
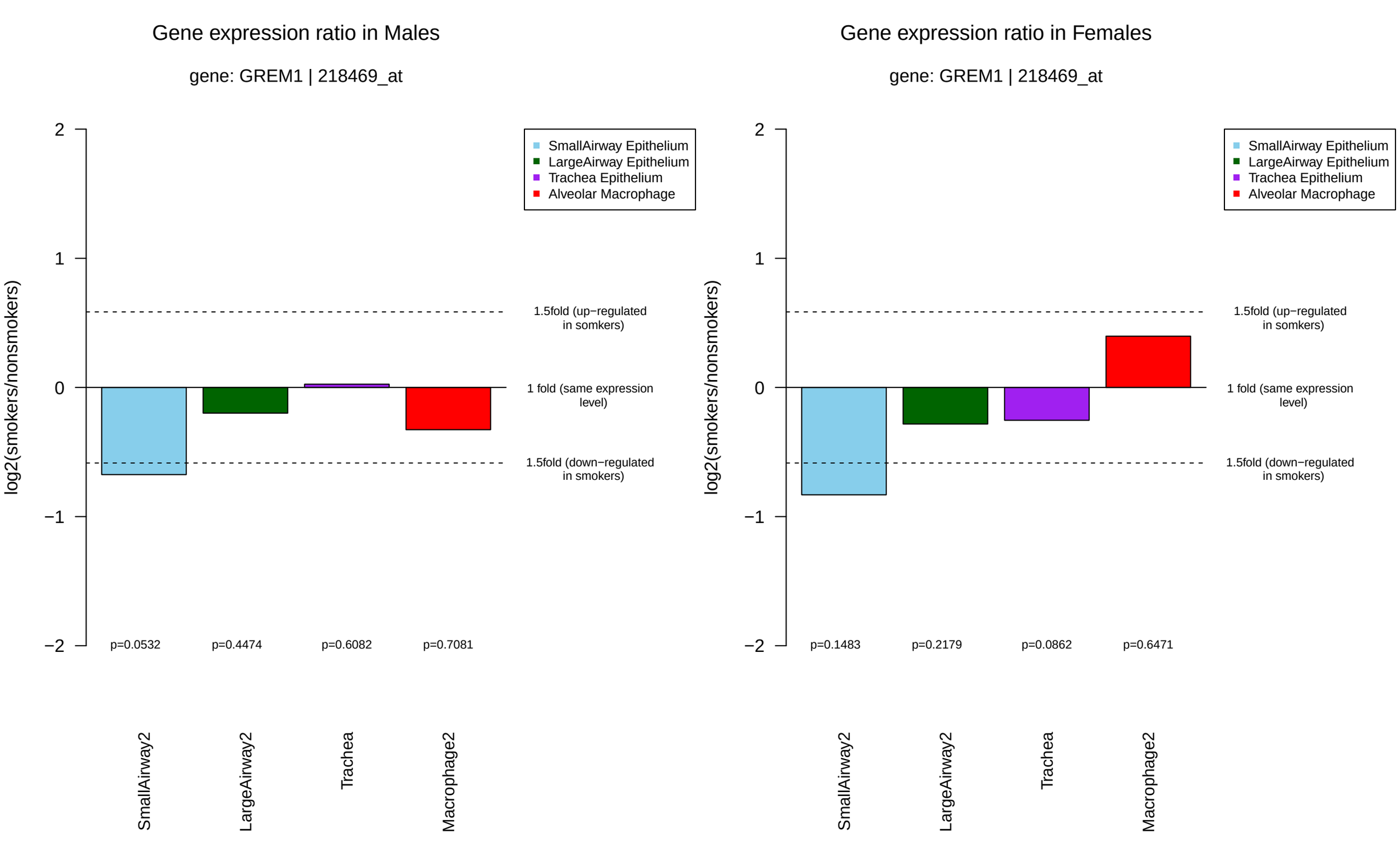
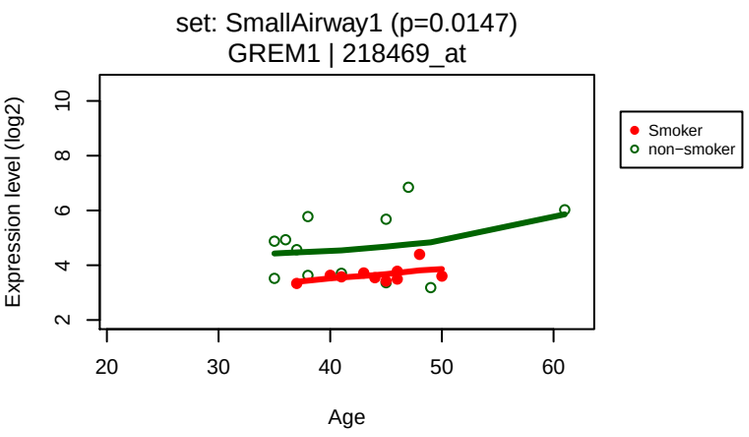
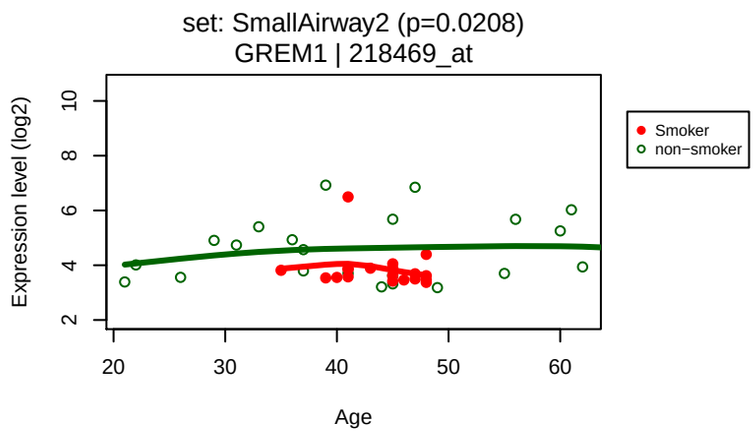
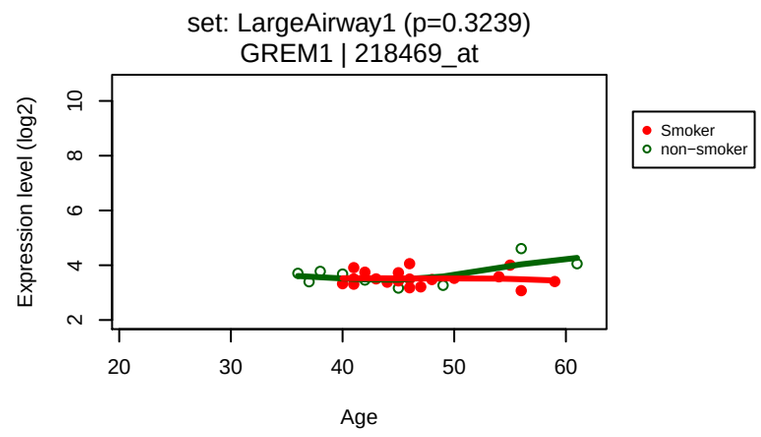

Gene expression of smokers and nonsmokers
gene: GREM1 | 218469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0147
p=0.0208
p=0.0436
p=0.3239
p=0.2725
p=0.3317
p=0.2929
p=0.9058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

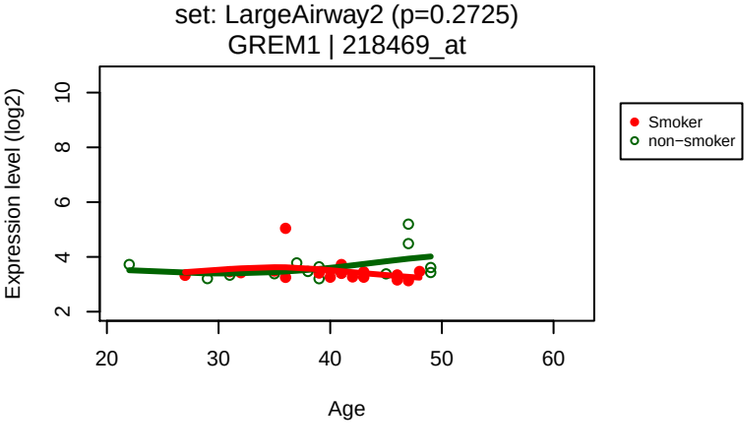

Gene expression ratio (smokers/nonsmokers)
gene: GREM1 | 218469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0147
p=0.0208
p=0.0436
p=0.3239
p=0.2725
p=0.3317
p=0.2929
p=0.9058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

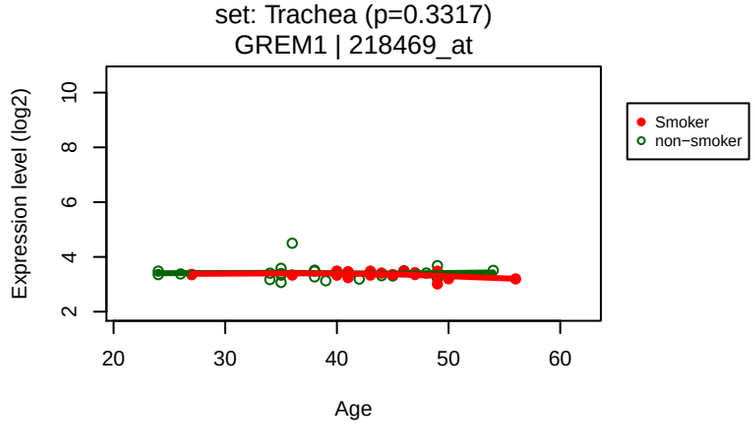

Gene expression ratio in Males
gene: GREM1 | 218469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0532
p=0.4474
p=0.6082
p=0.7081
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GREM1 | 218469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1483
p=0.2179
p=0.0862
p=0.6471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

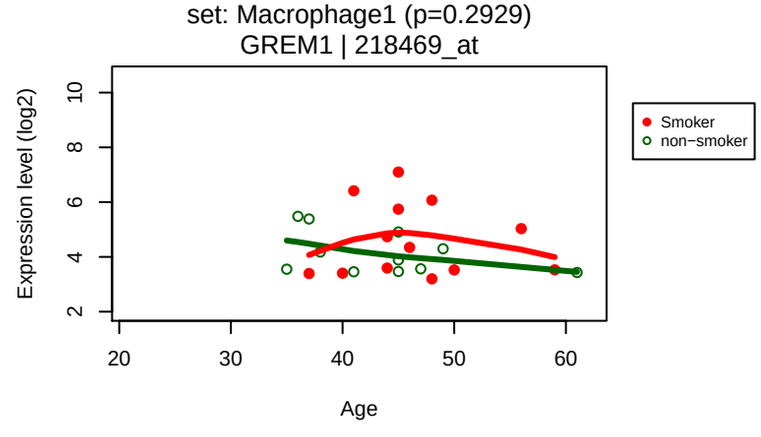

set: SmallAirway1 (p=0.0147)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker

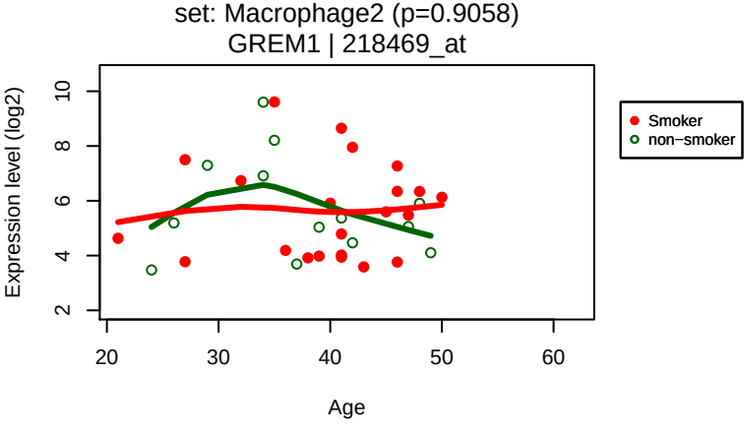

set: SmallAirway2 (p=0.0208)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker

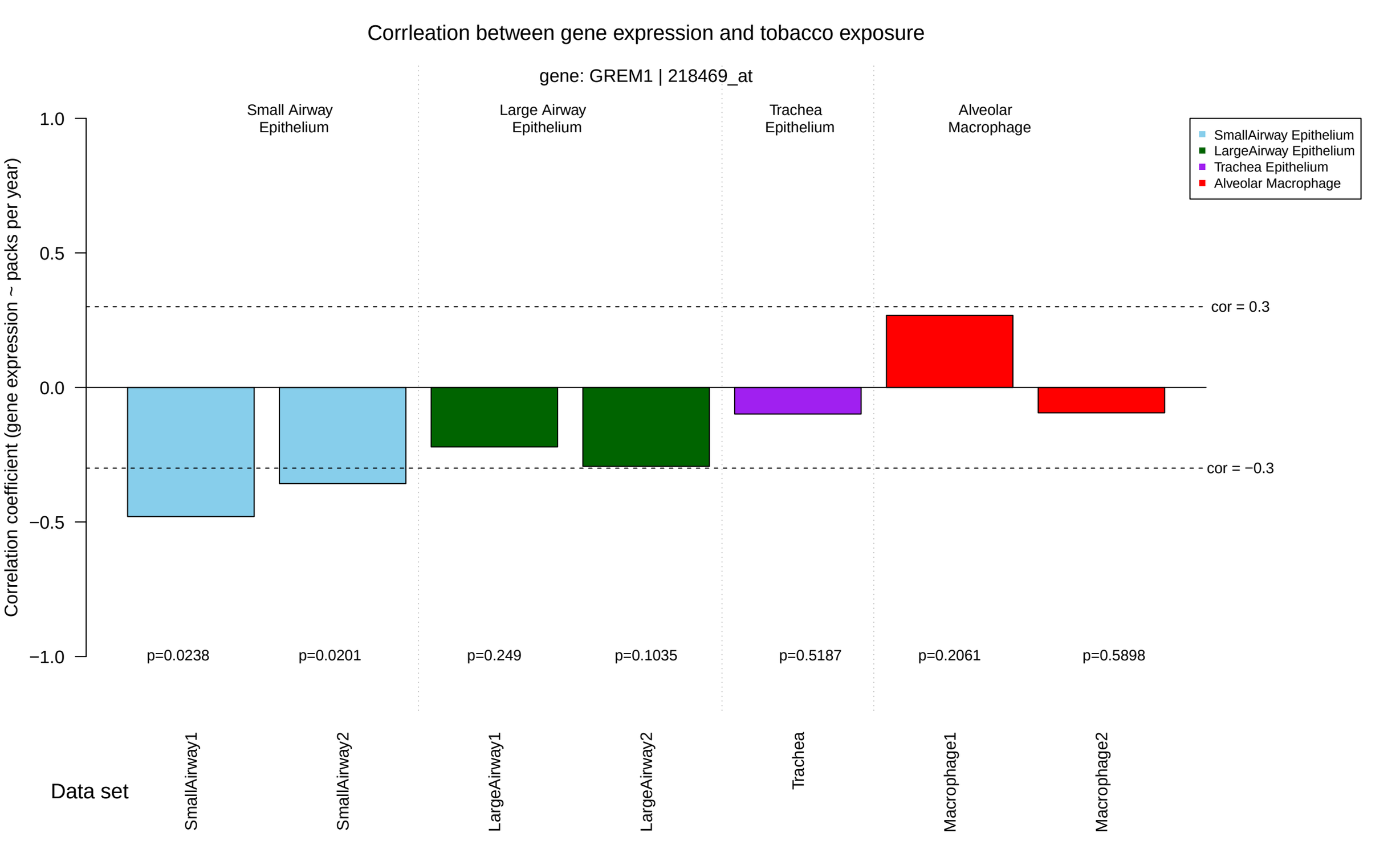

set: LargeAirway1 (p=0.3239)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2725)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3317)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2929)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9058)
GREM1 | 218469_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GREM1 | 218469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0238
p=0.0201
p=0.249
p=0.1035
p=0.5187
p=0.2061
p=0.5898
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2